Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels

Milestone 2: Data Loading & Data Modeling

Progress Report – Milestone 2

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1. Introduction

Data modeling plays a crucial role in transforming raw data into meaningful and structured information that can be used for analysis and decision-making. In business intelligence tools like Power BI, a well-designed data model improves performance, ensures accurate calculations, and makes report creation more efficient.

This documentation describes the data modeling process for the **Hotel Booking dataset**. The objective of this model is to organize booking-related data into a structured format that supports reporting and analytical insights.

The model has been designed using a **Star Schema architecture**, which is a widely accepted best practice in dimensional modeling. This structure separates transactional data from descriptive data, allowing better performance and simplified relationships.

2. Dataset Overview

The dataset used in this project contains hotel booking records, including details about customers, booking dates, hotels, market segments, and countries. The data was structured into fact and dimension tables to support analytical reporting.

The dataset includes the following key information:

- Booking arrival dates
- Hotel names
- Customer types

- Market segments
- Country information
- Other booking-related attributes

Before modeling, the data was cleaned and structured appropriately to ensure consistency and reliability.

3. Data Model Architecture

The data model follows a **Star Schema design**, where a central fact table is surrounded by multiple dimension tables. This structure simplifies querying and improves report performance.

In this model:

- The **Fact_HotelBookings** table acts as the central table.
- Dimension tables provide descriptive attributes for filtering and grouping.

The tables included in the model are:

- **Fact_HotelBookings**
- **DateTable**
- **Dim_Country**
- **Dim_CustomerType**
- **Dim_Hotel**
- **Dim_MarketSegment**

Each dimension table is connected directly to the fact table through a one-to-many relationship

DateTable

|

Dim_Hotel — Fact_HotelBookings — Dim_CustomerType

|

Dim_MarketSegment

|

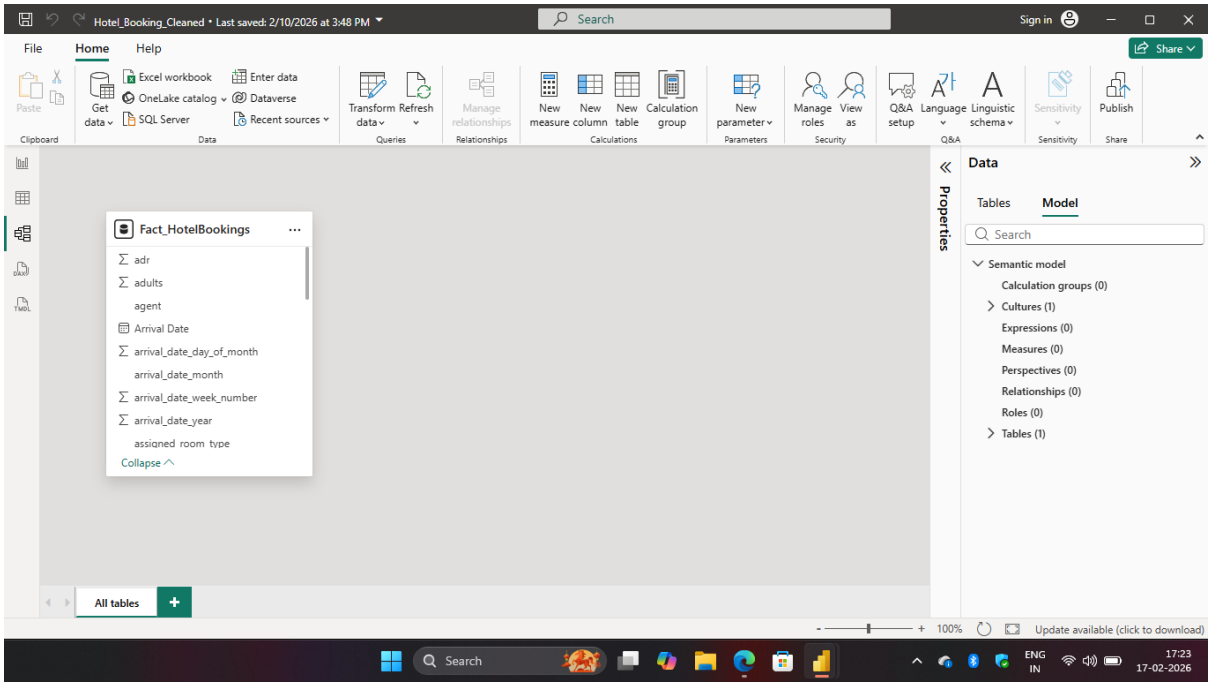
Dim_Country

4. Fact Table Description

The **Fact_HotelBookings** table contains the transactional data for hotel bookings. It serves as the core of the data model and stores measurable and analytical information.

This table includes:

- Arrival Date
- Country
- Hotel
- Customer Type
- Market Segment
- Booking-related metrics



Each record in this table represents an individual booking entry.

The fact table connects to dimension tables using foreign keys, allowing detailed analysis across multiple perspectives such as time, geography, and customer category.

5. Dimension Tables Description

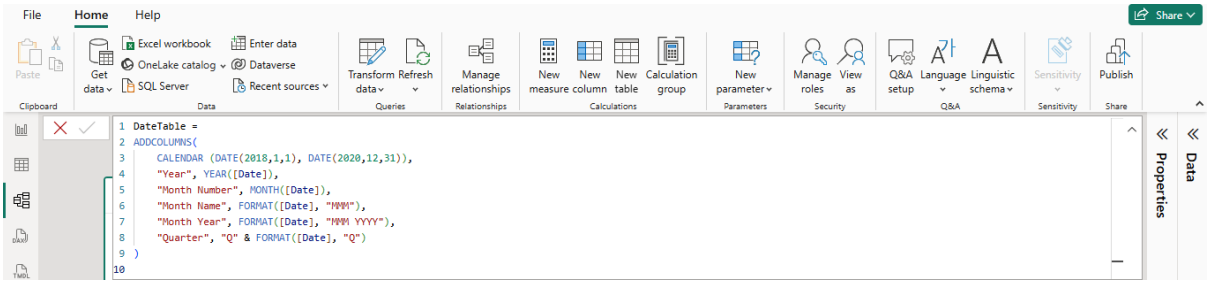
Dimension tables provide descriptive information that helps categorize and filter fact data. They contain unique values and are placed on the "one" side of relationships.

5.1 DateTable

The DateTable is used for time-based analysis and reporting. It enables grouping data by year, quarter, month, and other time attributes.

It includes:

- Date
- Month Name
- Month Number
- Month Year
- Quarter

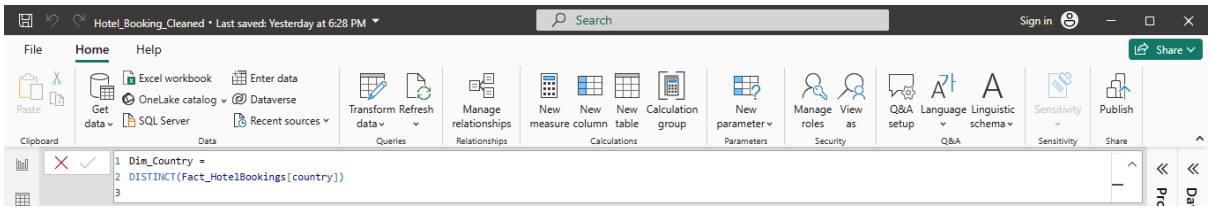


5.2 Dim_Country

The Dim_Country table stores unique country values used in booking records.

This table helps:

- Analyze bookings by country
- Compare regional performance
- Filter reports by geographic location

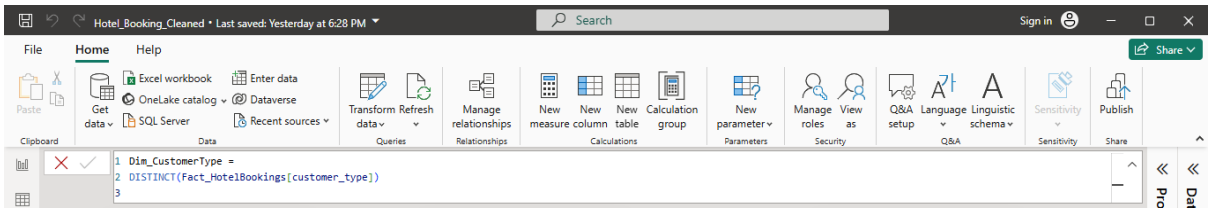


5.3 Dim_CustomerType

This table contains different categories of customers.

It helps in:

- Segmenting bookings by customer type
- Understanding customer behavior
- Comparing booking patterns

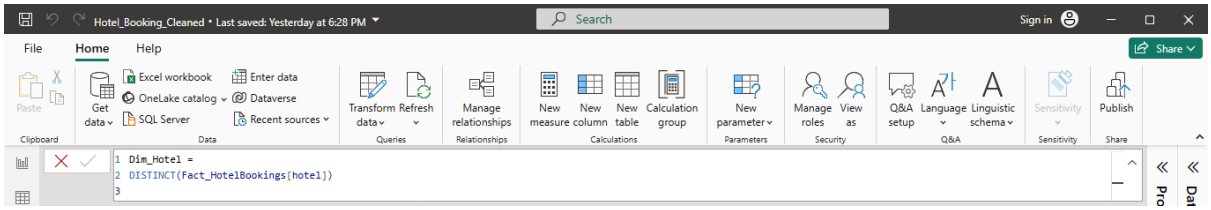


5.4 Dim_Hotel

The Dim_Hotel table includes unique hotel names.

It allows:

- Performance comparison between hotels
- Hotel-level filtering
- Revenue or booking distribution analysis



5.5 Dim_MarketSegment

This table stores market segmentation information.

It helps:

- Analyze booking channels
- Understand target markets
- Compare segment performance

6. Table Relationships

Relationships define how tables interact with each other. In this model, each dimension table is connected to the Fact_HotelBookings table using a one-to-many relationship.

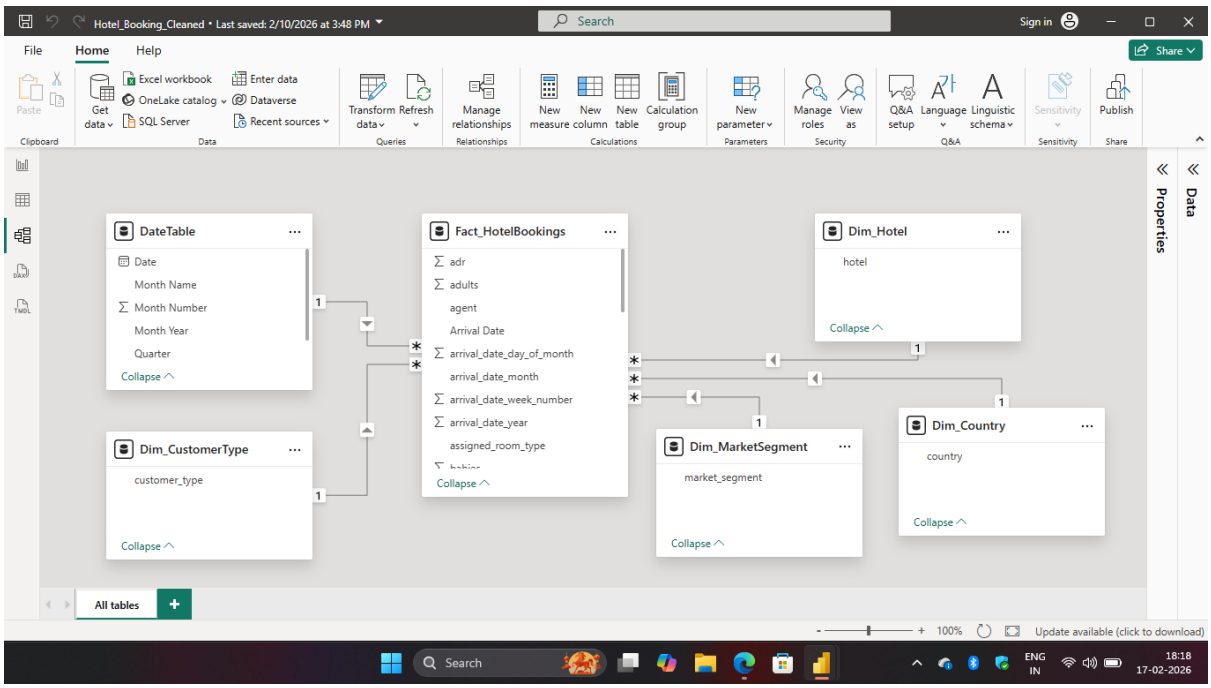
The relationships are defined as follows:

- DateTable[Date] → Fact_HotelBookings[Arrival Date]
- Dim_Country[Country] → Fact_HotelBookings[Country]
- Dim_CustomerType[Customer Type] → Fact_HotelBookings[Customer Type]
- Dim_Hotel[Hotel] → Fact_HotelBookings[Hotel]
- Dim_MarketSegment[Market Segment] → Fact_HotelBookings[Market Segment]

All relationships are:

- Active
- One-to-Many (1 → *)
- Single-direction filtering

Single-direction filtering ensures that dimension tables filter the fact table without causing ambiguity.



7. Cardinality and Filter Direction

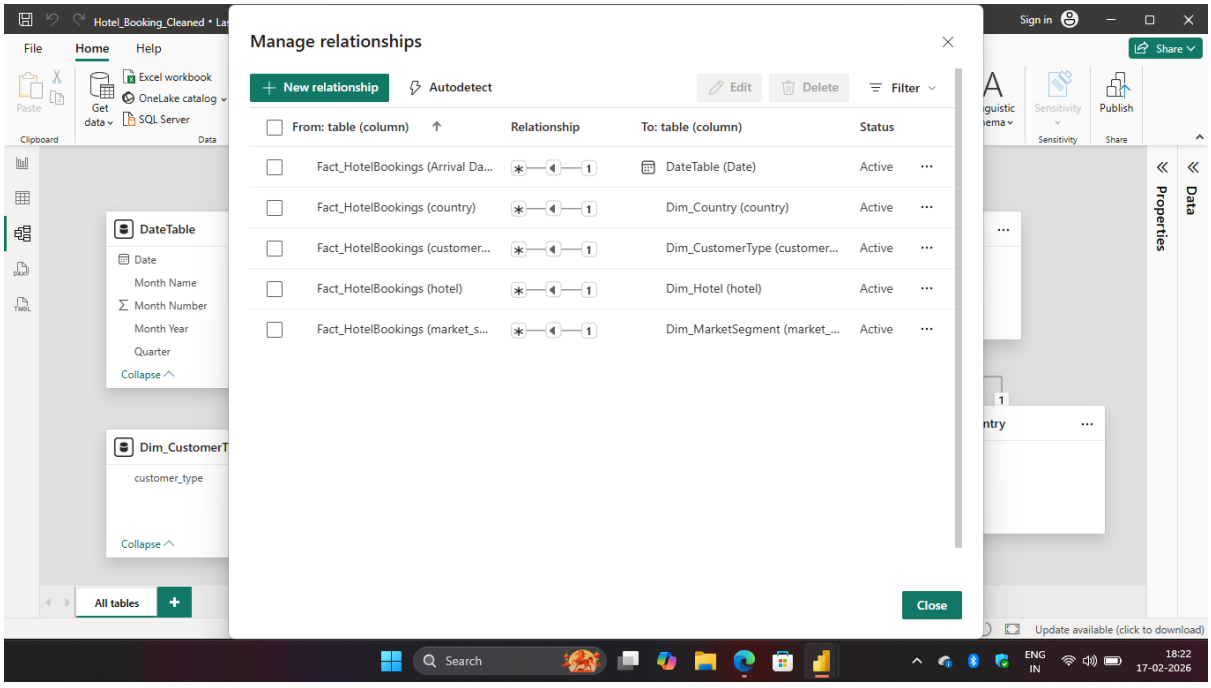
Cardinality defines how tables are related. In this model:

- Dimension tables represent the "One" side
- Fact table represents the "Many" side

This setup ensures:

- Data integrity
- Accurate aggregations
- Better performance

Filter direction is set to single direction from dimension to fact table. This prevents circular dependencies and ambiguous filtering behavior.



8. Benefits of the Data Model

The implemented Star Schema provides several advantages:

- Improved query performance
- Simplified report development
- Reduced redundancy
- Clear separation of transactional and descriptive data
- Better scalability

9. Conclusion

The Hotel Booking data model has been successfully designed using a Star Schema architecture. The separation of fact and dimension tables ensures clarity, performance efficiency, and analytical flexibility.

By implementing proper one-to-many relationships and maintaining single-direction filtering, the model supports reliable reporting and scalable dashboard development.

This structured approach makes the dataset suitable for advanced business intelligence analysis and future expansion.

Overall, the data modeling process has created a strong foundation for generating meaningful insights from hotel booking data.

Dashboard Title: Booking Behaviour Analysis — Hotel Revenue AI Dashboard

Introduction

This dashboard was developed using **Microsoft Power BI** to analyze customer booking behaviour in hotel reservation data from 2018 to 2020.
The purpose of this dashboard is to identify booking patterns, customer segments, cancellation trends, and guest loyalty insights to support data-driven decision making.

Objectives

The main objectives of the dashboard are:

- Analyze lead time patterns of bookings
- Identify dominant customer types
- Evaluate booking distribution channels
- Measure cancellation behaviour
- Compare repeat guests with new guests

Dataset Description

The dataset contains historical hotel booking records including:

- Hotel type
- Booking lead time
- Customer type
- Distribution channel
- Cancellation status
- Repeat guest indicator
- Arrival dates and stay details

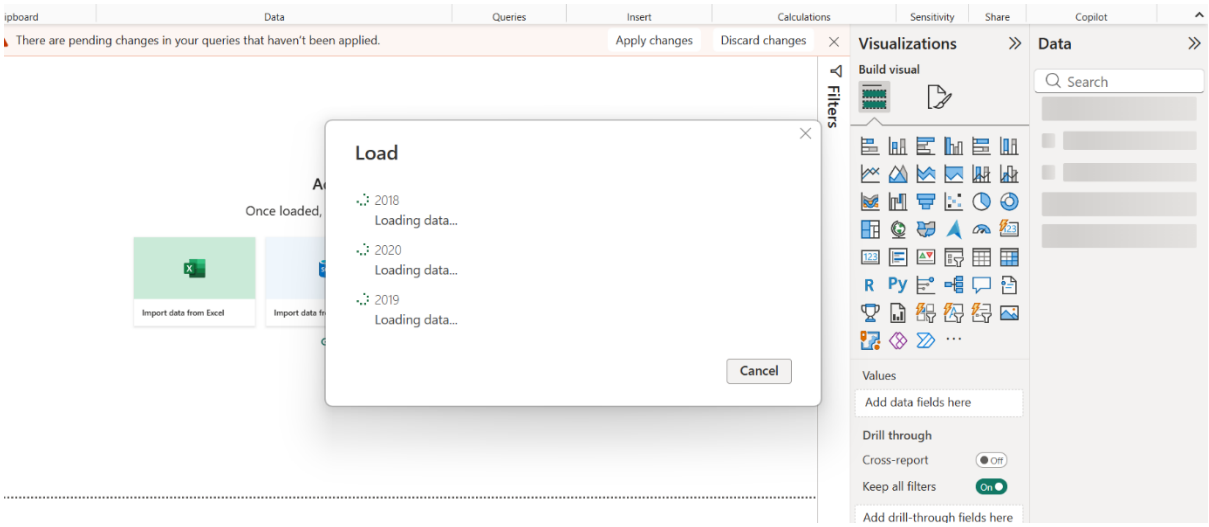
Data for the years **2018, 2019, and 2020** were used.

Data Loading Process

The dataset was imported into Power BI using:

Home → Get Data → Excel

Three tables (2018, 2019, and 2020) were loaded from the Excel file.



Data Modeling

Data modeling was performed to create a unified structure:

- Power Query Editor was opened using **Transform Data**
- Tables 2018, 2019, and 2020 were combined using **Append Queries**
- A single consolidated table named **HotelData** was created

This combined table acts as the **Fact Table**, containing all booking transactions used for analysis.

Basic preprocessing steps included:

- Header promotion
- Data type correction
- Handling minor row errors

Measures and Calculations

A KPI measure was created to calculate the cancellation rate:

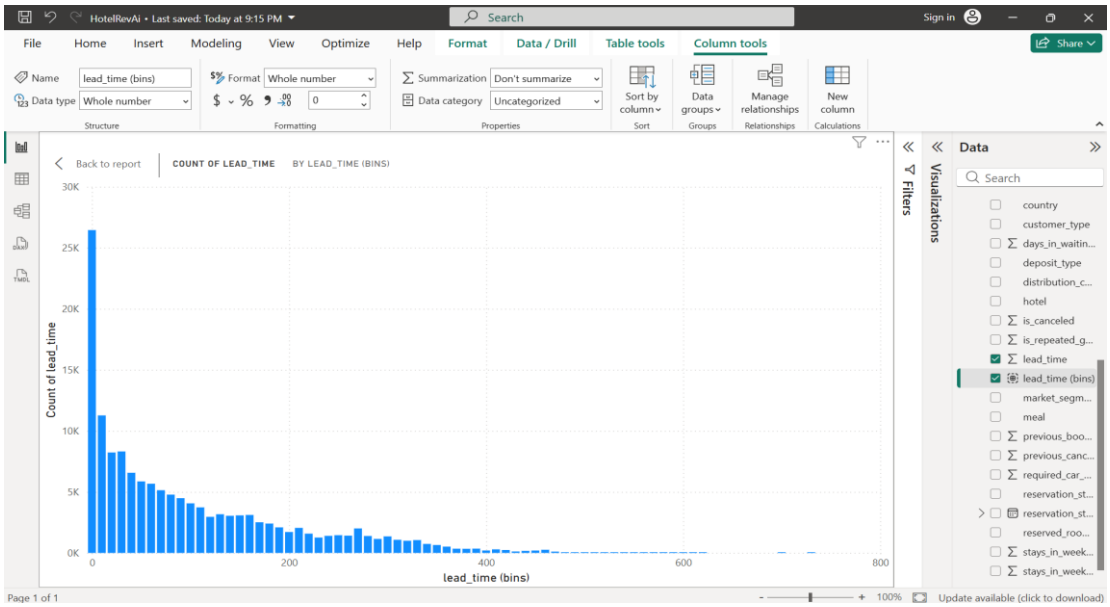
Cancellation Rate =
DIVIDE(
CALCULATE(COUNTROWS(HotelData), HotelData[is_canceled] = 1),
COUNTROWS(HotelData)
)

This measure displays the percentage of canceled bookings.

Dashboard Visualizations

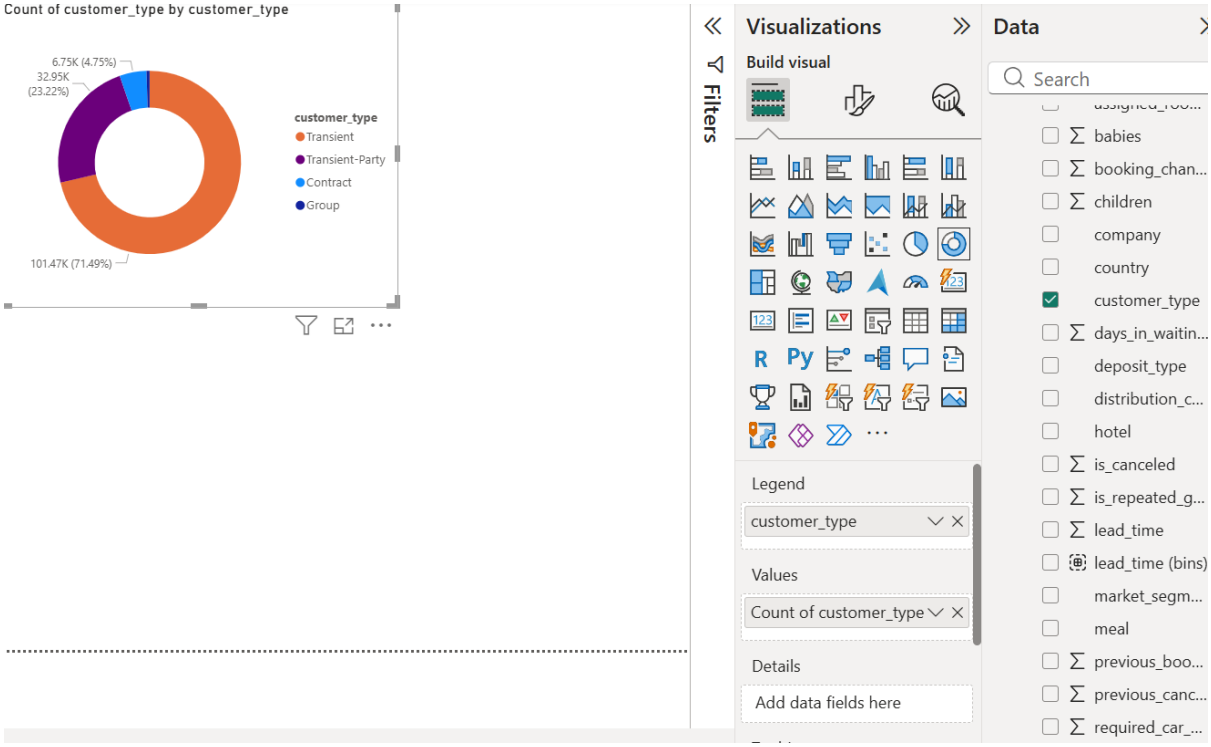
Lead Time Distribution

- Visual Type: Column Chart (Histogram using bins)
- Purpose: Shows how early customers make bookings.



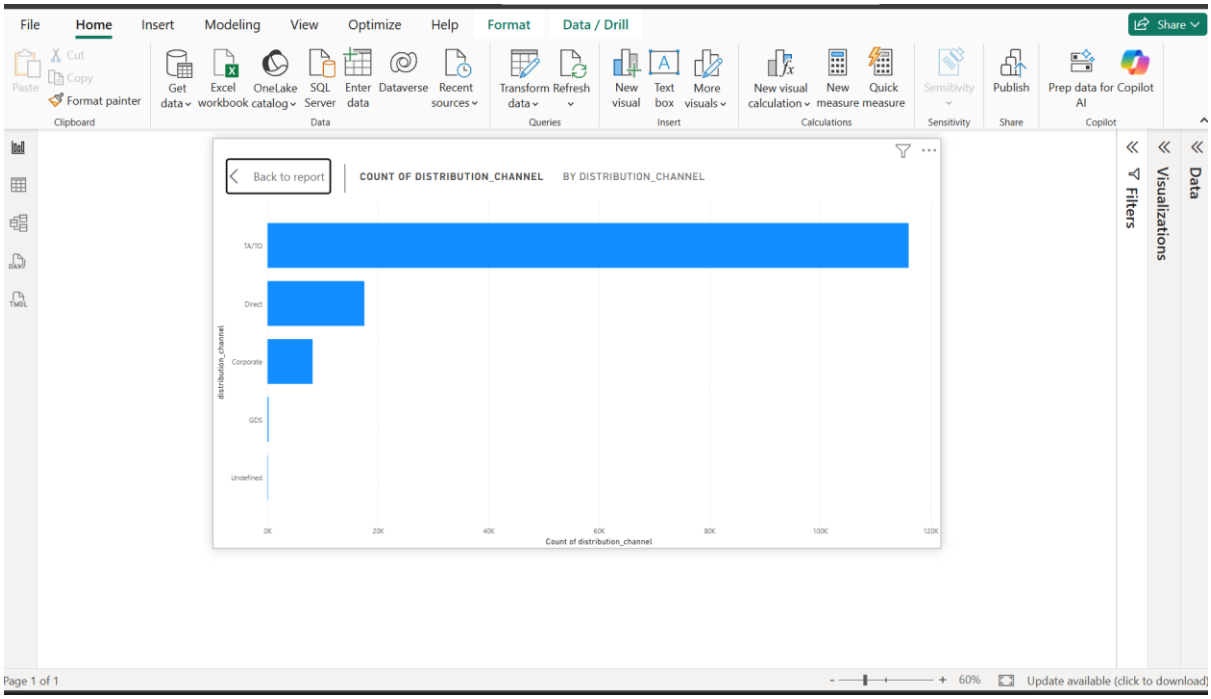
- Insight: Most bookings occur with shorter lead times.

Customer Type Distribution

- Visual Type: Donut Chart
- Field: customer_type
- Purpose: Displays booking distribution among customer segments.
- Insight: Transient customers form the majority.
- 

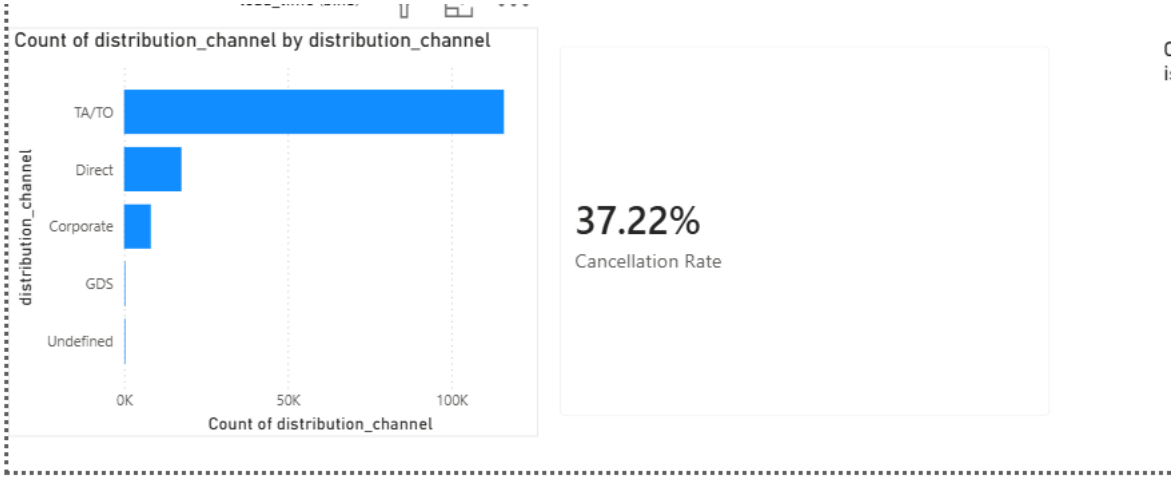
Booking by Channel

- Visual Type: Bar Chart
- Field: distribution_channel
- Purpose: Shows booking sources such as TA/TO, Direct, and Corporate.
- Insight: Travel agencies dominate booking channels.



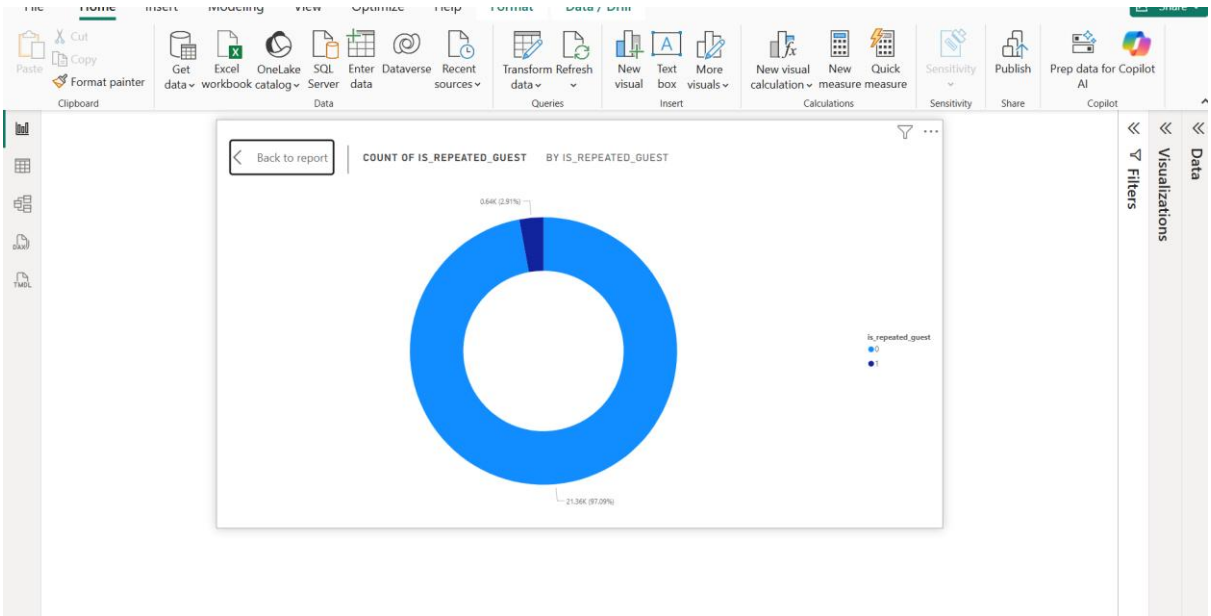
Cancellation Rate KPI

- Visual Type: Card
- Purpose: Displays overall cancellation percentage.
- Insight: Helps evaluate booking reliability and policy effectiveness.

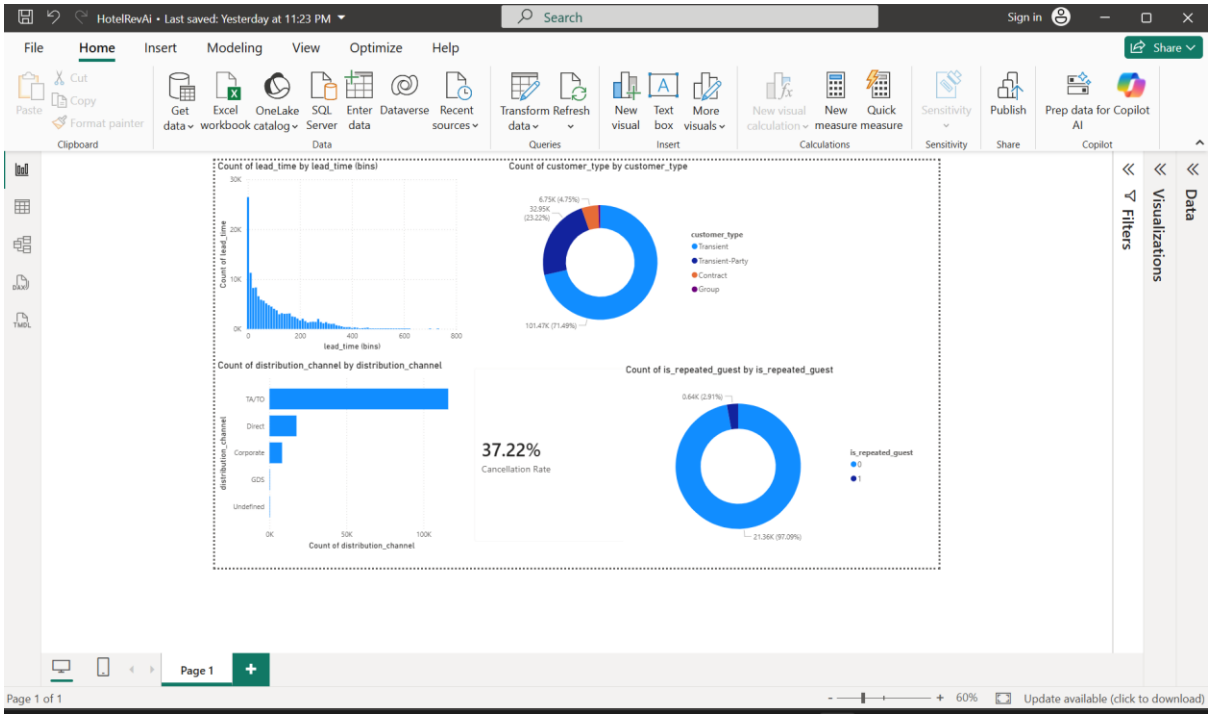


Repeat vs New Guests

- Visual Type: Donut Chart
- Field: is_repeated_guest
- Purpose: Compares new guests with returning guests.
- Insight: Majority of bookings are from new guests.



Final Dashboard



Key Insights

- Short lead time bookings are most common.
- Transient customers dominate reservations.
- Travel agencies contribute the highest booking volume.
- Cancellation rate is significant, indicating opportunity for policy optimization.
- Guest retention can be improved as repeat guests are fewer

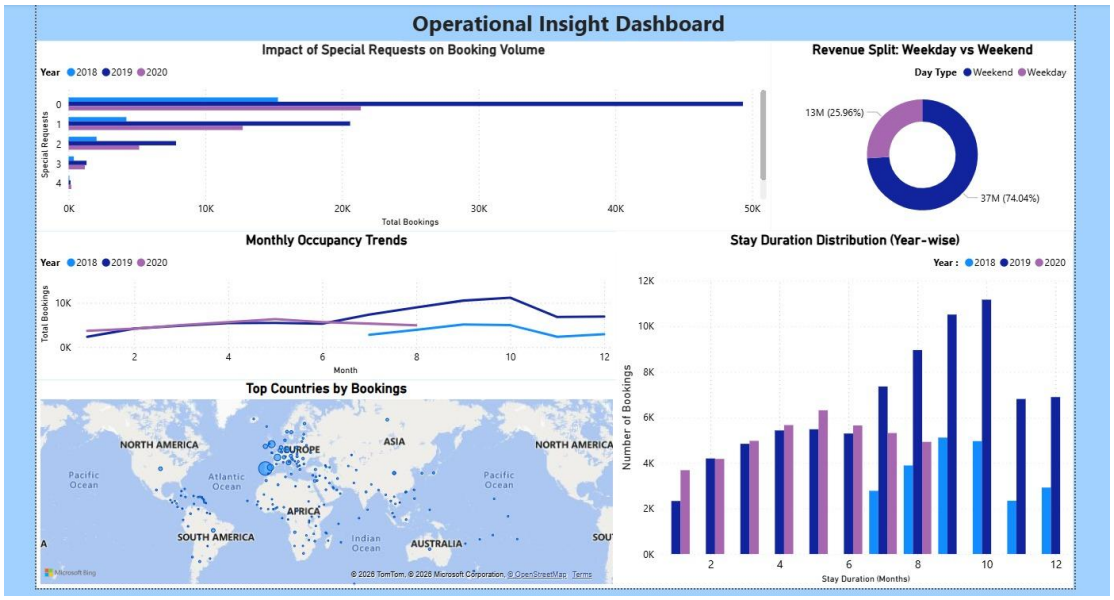
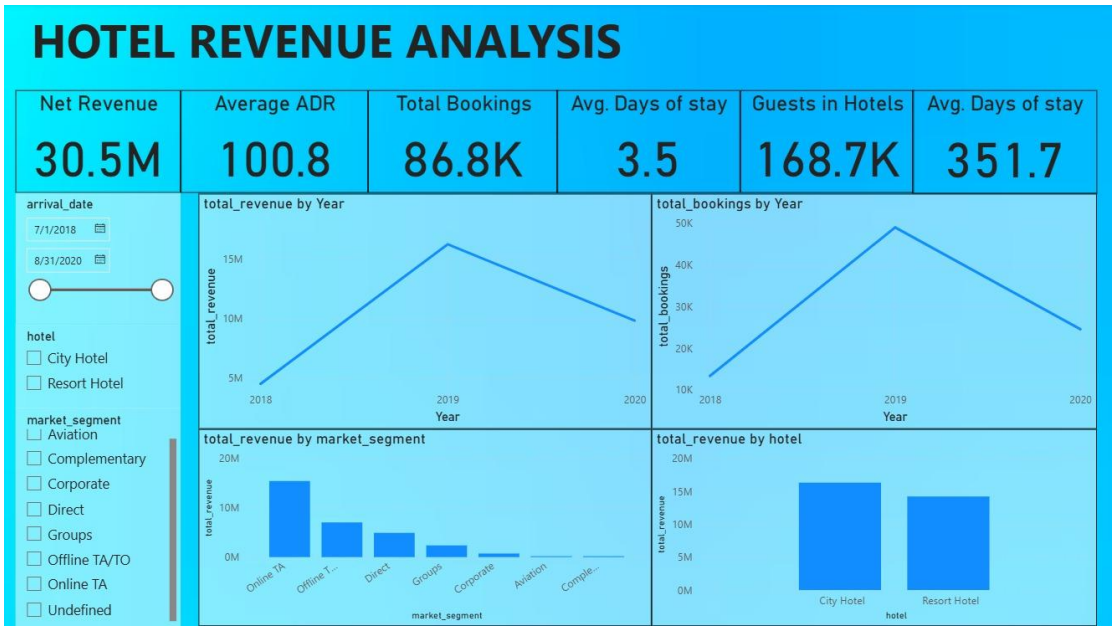
Tools and Technologies Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)

Conclusion

The Booking Behaviour Analysis dashboard provides a comprehensive view of customer booking patterns and operational performance.
By integrating multiple yearly datasets into a single fact table and applying interactive visualizations, the dashboard enables hotel management to make informed decisions related to marketing strategies, pricing, and customer retention.

Operation Insight Dashboard



Hotel Revenue Analysis